PADHAKU
Ganita Prakash · Class 6 Mathematics



CHAPTER 9

Symmetry

NCERT Exercise Solutions

Introduction	4 Q
9.1 Line of Symmetry · Figure it Out	2 Q
Figures with more than one line of symmetry	3 Q
Reflection	1 Q
Generating shapes having lines of symmetry	3 Q
Figure it Out · Punching Game and paper cutting	17 Q
9.2 Rotational Symmetry	4 Q
Rotational Symmetry of Figures with Radial Arms	8 Q
Figure it Out · 9.2 Rotational Symmetry	6 Q
Symmetries of a circle	1 Q
Figure it Out · Symmetries of a circle	14 Q
Playing with Tiles	3 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
Independent study only. Not affiliated with or endorsed by the official NCERT publication.

⭐ Introduction

? **What else about the butterfly appeals to you?**

> 💡 **SAMPLE ANSWER**
>
> Its left wing and right wing are exactly the same shape and pattern.
>
> One half is a **mirror image** of the other half.

? **Can you see what repeats in the beautiful rangoli figure?**

> ✔ **ANSWER**
>
> The same petal design repeats **4 times** around the centre.
>
> When the rangoli is turned by 90° about the centre, every part comes back onto a matching part.

? **What about the pinwheel? Can you spot which pattern is repeating? (Hint: look at the hexagon first.)**

What figure repeats along each side of the hexagon? What is its shape? How do these shapes move as you go along the boundary?

> 💡 **SAMPLE ANSWER**
>
> The same **triangle** is stuck on each of the 6 sides of the hexagon.

As we go around the hexagon, the triangle repeats in the same way, turning along with the side (like a turning wheel).

(check the picture in your book)

? **What are the symmetries that you see in the Taj Mahal and the Gopuram?**

💡 **SAMPLE ANSWER**

Both have a **vertical line of symmetry** through the middle: the left half is a mirror image of the right half.

Domes, minarets, arches and carvings repeat in a regular pattern.

🧮 9.1 Line of Symmetry · Figure it Out

1 **Do you see any line of symmetry in the figures at the start of the chapter? What about in the picture of the cloud?**

✔ **ANSWER**

- Flower: yes, many lines of symmetry (6).
- Butterfly: yes, **1** (vertical line through its body).
- Rangoli: yes, **4**.
- Pinwheel: **no** line of symmetry (it only looks the same when turned).
- Cloud: **no** line of symmetry.

2 **For each of the following figures, identify the line(s) of symmetry if it exists.**

💡 **SAMPLE ANSWER**

For each figure, imagine folding it along a line.

If the two parts overlap exactly, that line is a line of symmetry; draw it as a dotted line.

If no fold makes the halves overlap, the figure has **no** line of symmetry.

(check the picture in your book)

🧮 Figures with more than one line of symmetry

? **Is there any other way to fold the square so that the two halves overlap? How many lines of symmetry does the square shape have?**

✔ **ANSWER**

No, there is no other way.

Lines: 1 vertical, 1 horizontal, 2 diagonals.

A square has **4 lines of symmetry**.

? **The figures below also have multiple lines of symmetry. Can you find them all?**

💡 **SAMPLE ANSWER**

Fold (or imagine folding) each figure in every possible way: vertically, horizontally and slantwise.

Every fold where both halves overlap exactly is a line of symmetry; count them all.

(check the picture in your book)

? **Take a rectangle that is not a square. Is its diagonal a line of symmetry? Fold a rectangular paper along its diagonal. What do you observe?**

✔ ANSWER

The two triangles do **not** overlap exactly; corners stick out.

So the diagonal of a rectangle (not a square) is **not a line of symmetry**.

A rectangle has only 2 lines of symmetry (vertical and horizontal).

⭐ Reflection

? **Square ABCD (A top-left, B top-right, C bottom-right, D bottom-left).**
What if we reflect along the diagonal from A to C? Where do points A, B, C and D go? What if we reflect along the horizontal line of symmetry?

✔ ANSWER

Along diagonal AC: A and C stay where they are; B goes to D's place and D goes to B's place.

Along the horizontal line: A goes to D's place, D to A's place, B to C's place and C to B's place.

✏️ Generating shapes having lines of symmetry

Ink Blot **Fold a paper, put ink drops on one half, press the halves together and open.**

What do you see? Is the figure symmetric? If yes, where is the line of symmetry? Is there any other line along which it can be folded into two identical parts?

💡 SAMPLE ANSWER

We see an ink pattern on both halves; the two halves are mirror images.

Yes, the figure is **symmetric**.

The line of symmetry is the **fold line**.

Usually there is **no other** such line, because the ink spreads in a random shape.

Paper cut **A folded sheet is cut along the dotted line. Draw a sketch of how the paper will look when unfolded. Do you see a line of symmetry in this figure? What is it?**

💡 SAMPLE ANSWER

When unfolded, the cut shape appears on both sides of the fold, as mirror images.

Yes, the **fold line** is the line of symmetry.

(check the picture in your book)

Paper strip **Fold thin rectangular paper several times, cut patterns and identify the lines of symmetry in the repeating design.**

💡 SAMPLE ANSWER

Every fold line becomes a line of symmetry of the repeating design.

So the number of lines of symmetry = number of folds made in the strip (the vertical creases).

🧮 Figure it Out · Punching Game and paper cutting

Game **Punch holes at different locations of a folded square sheet of paper and create different symmetric patterns.**

💡 SAMPLE ANSWER

Each hole punched on the folded sheet makes a matching hole on the other side of the fold.

So every pattern is symmetric about the fold line.

1 **In each figure, a hole was punched in a folded square sheet and then unfolded. Identify the line along which the paper was folded. Figure (d) was created by punching a single hole. How was the paper folded?**

💡 **SAMPLE ANSWER**

(a)–(c): the fold line is the line exactly midway between the two holes (each hole is the mirror image of the other).

(d): it has 4 holes from a single punch, so the paper was folded **twice**: first vertically, then horizontally (or the other way round).

(check the picture in your book)

2 **Given the line(s) of symmetry, find the other hole(s).**

💡 **SAMPLE ANSWER**

For each hole, measure its distance from the line of symmetry.

Mark the other hole at the same distance on the opposite side, straight across the line.

With two lines of symmetry, reflect in both lines: 1 hole becomes 4 holes.

(check the picture in your book)

3 **Here are some questions on paper cutting. A vertical fold and a horizontal fold are shown by symbols.**

✔ ANSWER

This explains the symbols used in Q4: a vertical fold folds the paper left-right, a horizontal fold folds it top-bottom.

4 **After each of the following cuts, predict the shape of the hole when the paper is opened. Then make the cutouts and verify.**

💡 SAMPLE ANSWER

Rule: every cut is reflected in each fold.

- One fold: the cut shape and its mirror image join along the fold.
- Two folds (vertical + horizontal): the cut repeats 4 times, so a cut at the folded corner gives a shape in the centre of the paper.

(check the picture in your book)

5 (a) **Get this shape with some folds and a single straight cut: the hole in the centre is a square (sides parallel to the edges).**

💡 SAMPLE ANSWER

Fold the paper in half, then in half again (the centre is now the closed corner).

Fold once more along the diagonal through the closed corner.

Make one straight cut **at right angles to a folded edge**, close to the corner.

Open: a square hole appears in the centre.

(check the picture in your book)

5 (b) **Get this shape with some folds and a single straight cut: the hole in the centre is a square (tilted, like a diamond).**

💡 **SAMPLE ANSWER**

Fold the paper in half horizontally, then vertically.

At the closed corner (centre), make one slanting straight cut, cutting off **equal lengths** on both folded edges.

Open: a square hole (standing on a corner) appears in the centre.

(check the picture in your book)

6 (a) **How many lines of symmetry does this shape have?**

💡 **SAMPLE ANSWER**

Count all the fold lines where the two halves overlap exactly.

(check the picture in your book)

6 (b) **A triangle with equal sides and equal angles.**

✔ ANSWER

One line from each corner to the middle of the opposite side.

3 lines of symmetry.

6 (c) **A hexagon with equal sides and equal angles.**

✔ ANSWER

3 lines join opposite corners; 3 lines join middles of opposite sides.

6 lines of symmetry.

7 **Trace each figure and draw the lines of symmetry, if any.**

💡 SAMPLE ANSWER

Trace the figure, then fold the tracing in different ways.

Draw a dotted line along every fold where the halves overlap exactly; write "none" if no such fold exists.

(check the picture in your book)

8 **Find the lines of symmetry for the kolam below.**

💡 **SAMPLE ANSWER**

Check the vertical, horizontal and both diagonal lines through the centre of the kolam.

A square-shaped kolam with the same design in all directions has **4 lines of symmetry** (vertical, horizontal and 2 diagonals).

(check the picture in your book)

9 **Draw: (a) a triangle with exactly one line of symmetry, (b) a triangle with exactly three lines of symmetry, (c) a triangle with no line of symmetry.**

Is it possible to draw a triangle with exactly two lines of symmetry?

✔ **ANSWER**

(a) A triangle with only two equal sides (isosceles), e.g. sides 5 cm, 5 cm, 3 cm. The line from the top corner to the middle of the base is its line of symmetry.

(b) A triangle with all sides equal (equilateral), e.g. 4 cm, 4 cm, 4 cm.

(c) A triangle with all sides different (scalene), e.g. 3 cm, 4 cm, 5 cm.

No. If a triangle has two lines of symmetry, two pairs of sides are equal, so all three sides are equal; then it has 3 lines, not 2.

10 **Draw a figure with at least one curved boundary and (a) exactly one line of symmetry, (b) exactly two lines of symmetry, (c) exactly four lines of symmetry.**

💡 SAMPLE ANSWER

(a) A semicircle or a heart shape: 1 line.

(b) An oval (ellipse) or an eye shape: 2 lines (along its length and its width).

(c) A square with a semicircle drawn outward on each side, or a flower with 4 equal petals: 4 lines.

11 **Copy on squared paper and complete each figure so that the blue line is a line of symmetry. (Hint: for (c) and (f), rotate the book.)**

💡 SAMPLE ANSWER

For every corner point, count the squares to the blue line.

Mark a new point the same number of squares on the other side, straight across the line.

Join the new points in the same order. For slanting blue lines, (c) and (f), turn the book so the line looks straight up.

(check the picture in your book)

12 **Copy on squared paper and complete each drawing so that both blue lines are lines of**

symmetry.

💡 SAMPLE ANSWER

First reflect the given part in one blue line (count squares to the line).

Then reflect the whole new figure in the other blue line.

The finished figure has the same part in all four corners.

(check the picture in your book)

13 **Copy on a dot grid. For each figure draw two more lines to make a shape that has a line of symmetry.**

💡 SAMPLE ANSWER

Choose a line of symmetry (for example through the middle of the given lines).

Draw the two new lines as mirror images of the given lines across it, so the shape closes up.

(check the picture in your book)

✏️ 9.2 Rotational Symmetry

? **Will the windmill look exactly the same when rotated through an angle of less than 90°?**

✔ ANSWER

No. Its smallest angle of symmetry is 90°.

? **Do you know of any other shape that has exactly four angles of symmetry?**

✔ ANSWER

Yes. A **square**, or a plus sign (+) with equal arms.

Angles: 90°, 180°, 270°, 360°.

? **How many angles of symmetry does a square have? How much rotation does it require to get the initial square? Where is the centre of rotation?**

✔ ANSWER

It needs a rotation of **90°** to look the same again.

Angles of symmetry: 90°, 180°, 270°, 360°, so **4**.

The centre of rotation is where the **two diagonals meet**.

Example **A rotation of 180° of the strip gives a new position. Does this overlap with the original figure? Why?**

💡 SAMPLE ANSWER

No. After a half turn the parts of the strip are in the wrong places, so it does not match the original.

It matches only after 360°, so the strip has no rotational symmetry.

(check the picture in your book)

Rotational Symmetry of Figures with Radial Arms

? **A figure with 4 radial arms (90° between adjacent dotted lines). How many angles of symmetry does it have? What are they?**

✔ ANSWER

360° ÷ 4 = 90°

Angles: 90°, 180°, 270°, 360°

4 angles of symmetry.

? **Can you change the angles between the radial arms so that the figure still has 4 angles of symmetry?**

✔ ANSWER

No. Each quarter turn must bring every arm onto the next arm.

So all 4 angles must be equal: 360° ÷ 4 = 90°.

? **How will you modify the figure so that it has only two angles of symmetry?**

💡 SAMPLE ANSWER

Make only opposite angles equal, e.g. angles 60°, 120°, 60°, 120° between the dotted lines.

Now only a half turn matches it: angles **180° and 360°**.

? **Can we get a figure having exactly 3 angles of symmetry? Can you use radial arms for this?**

✔ ANSWER

Yes. Use 3 radial arms with equal angles between them.

? **The given figure with three radial arms: how many angles of symmetry does it have?**

✔ ANSWER

Its angles between arms are not equal, so only a full turn brings it back.

Only **360°**, so it has no rotational symmetry.

? **If ∠A = ∠B = ∠C, what must this angle be? How many angles of rotation does the figure then have and what are they?**

✔ ANSWER

360° ÷ 3 = **120°** each.

Angles of symmetry: 120°, 240°, 360°, so **3**.

? **Can you draw a figure with radial arms that has (a) exactly 5 angles of symmetry, (b) 6**

angles of symmetry? Find the angles of symmetry in each case.

✔ ANSWER

(a) Use 5 equal arms: 360° ÷ 5 = 72° between adjacent arms.

Angles: **72°, 144°, 216°, 288°, 360°**.

(b) Use 6 equal arms: 360° ÷ 6 = 60° between adjacent arms.

Angles: **60°, 120°, 180°, 240°, 300°, 360°**.

? **A figure with radial arms has exactly 7 angles of symmetry. What is its smallest angle of symmetry? Is it a whole number of degrees? If not, express it as a mixed fraction.**

✔ ANSWER

360° ÷ 7 = 51 remainder 3

Smallest angle = **51 3/7°**

No, it is not a whole number.

Figure it Out · 9.2 Rotational Symmetry

1 **Find the angles of symmetry for the given figures about the point marked •.**

✔ ANSWER

(a) **90°, 180°, 270°, 360°**

(b) **360°** only

(c) **180°, 360°**

2 **Which of the following figures have more than one angle of symmetry?**

💡 **SAMPLE ANSWER**

Turn each figure about its centre. If it looks the same before a full turn (e.g. at 90°, 120° or 180°), it has more than one angle of symmetry.

Figures that match only at 360° have just one angle of symmetry.

(check the picture in your book)

3 **Give the order of rotational symmetry for each figure.**

💡 **SAMPLE ANSWER**

Order = number of times the figure looks the same in one full turn.

Order = 360° ÷ smallest angle of symmetry (e.g. smallest angle 90° gives order 4).

(check the picture in your book)

? **In each case, the angles are the multiples of the smallest angle. Will this always happen? What do you think?**

✔ **ANSWER**

Yes. Turning by the smallest angle again and again gives the figure back each time.

So every angle of symmetry is a multiple of the smallest one.

T/F 1 **Every figure will have 360 degrees as an angle of symmetry.**

✔ ANSWER

True. A full turn always brings any figure back to its starting position.

T/F 2 **If the smallest angle of symmetry of a figure is a natural number in degrees, then it is a factor of 360.**

✔ ANSWER

True. Turning by the smallest angle again and again must reach exactly 360°, so it divides 360.

⭐ Symmetries of a circle

? **Find other objects around us having rotational symmetry (like wheels).**

💡 SAMPLE ANSWER

Ceiling fan, bicycle wheel, flowers, coins, a clock face without numbers, a pizza cut into equal slices, a car wheel cover.

🧮 Figure it Out · Symmetries of a circle

1 **Colour the sectors of the circle (12 equal sectors) so that the figure has (i) 3 angles of**

symmetry, (ii) 4 angles of symmetry. (iii) What are the possible numbers of angles of symmetry you can get by colouring the sectors in different ways?

💡 SAMPLE ANSWER

(i) Colour every 4th sector (sectors 1, 5, 9): angles 120°, 240°, 360°.

(ii) Colour every 3rd sector (sectors 1, 4, 7, 10): angles 90°, 180°, 270°, 360°.

(iii) The number must divide 12: **1, 2, 3, 4, 6 or 12**.

(check the picture in your book)

2 **Draw two figures other than a circle and a square that have both reflection symmetry and rotational symmetry.**

💡 SAMPLE ANSWER

- Equilateral triangle: 3 lines; angles 120°, 240°, 360°.
- Regular hexagon: 6 lines; angles 60°, 120°, ..., 360°.

(A rectangle also works: 2 lines; angles 180°, 360°.)

3 (a) **A triangle with at least two lines of symmetry and at least two angles of symmetry.**

✔ ANSWER

An **equilateral triangle**: 3 lines of symmetry; angles 120°, 240°, 360°.

3 (b) **A triangle with only one line of symmetry but not having rotational symmetry.**

✔ ANSWER

An **isosceles triangle** (only two sides equal): 1 line of symmetry; only 360°.

3 (c) **A quadrilateral with rotational symmetry but no reflection symmetry.**

✔ ANSWER

A **parallelogram** that is not a rectangle or rhombus: no line of symmetry; angles 180°, 360°.

3 (d) **A quadrilateral with reflection symmetry but not having rotational symmetry.**

✔ ANSWER

A **kite** or an isosceles trapezium: 1 line of symmetry; only 360°.

4 **In a figure, 60° is the smallest angle of symmetry. What are the other angles of symmetry?**

✔ ANSWER

Multiples of 60° up to 360°:

120°, 180°, 240°, 300°, 360°

5 **In a figure, 60° is an angle of symmetry. The figure has two angles of symmetry less than 60°. What is its smallest angle of symmetry?**

✔ ANSWER

Let the smallest angle be x. Angles of symmetry are x, 2x, 3x, ...

Two angles less than 60° are x and 2x, so 3x = 60°.

x = 60° ÷ 3 = **20°**

6 **Can we have a figure with rotational symmetry whose smallest angle of symmetry is (a) 45°? (b) 17°?**

✔ ANSWER

(a) 360 ÷ 45 = 8. **Yes** (45 is a factor of 360; e.g. 8 equal radial arms).

(b) 360 ÷ 17 = 21 remainder 3. **No** (17 is not a factor of 360).

7 **Picture of the new Parliament Building in Delhi (triangular outer boundary).**

(a) Does the outer boundary have reflection symmetry? If so, how many lines? (b) Does it have rotational symmetry around its centre? If so, find the angles.

✔ ANSWER

(a) **Yes**, **3 lines of symmetry**, one from each corner to the middle of the opposite side.

(b) **Yes**. Angles: **120°, 240°, 360°**.

8 **How many lines of symmetry do the shapes in the Regular Polygons sequence (Chapter 1, Table 3) have? What number sequence do you get?**

✔ ANSWER

Shape	Lines of symmetry
Triangle	3
Square	4
Pentagon	5
Hexagon	6
Heptagon	7
Octagon	8
Nonagon	9
Decagon	10

Sequence: **3, 4, 5, 6, 7, 8, 9, 10** (counting numbers from 3).

9 **How many angles of symmetry do the shapes in the Regular Polygons sequence have? What number sequence do you get?**

✔ ANSWER

A regular polygon with n sides has n angles of symmetry (smallest angle 360° ÷ n).

Sequence: **3, 4, 5, 6, 7, 8, 9, 10**.

10 **How many lines of symmetry and angles of symmetry do the shapes in the Koch Snowflake sequence (Chapter 1, Table 3) have?**

✔ ANSWER

First shape (triangle): **3 lines**, **3 angles** (120°, 240°, 360°).

Every later snowflake: **6 lines**, **6 angles** (60°, 120°, ..., 360°).

Lines: 3, 6, 6, 6, 6; angles: 3, 6, 6, 6, 6.

11 **How many lines of symmetry and angles of symmetry does the Ashoka Chakra have?**

✔ ANSWER

It has 24 equal spokes.

24 lines of symmetry and **24 angles of symmetry** (smallest angle 360° ÷ 24 = 15°).

⭐ Playing with Tiles

a **Use the colour tiles to complete the given figure so that it has exactly 2 lines of symmetry.**

💡 SAMPLE ANSWER

Place tiles so the left half mirrors the right half and the top half mirrors the bottom half.

Avoid making the diagonals lines of symmetry too.

(check the picture in your book)

b **Use 16 tiles to make figures that have exactly (i) 1 line of symmetry, (ii) 2 lines of symmetry.**

💡 SAMPLE ANSWER

(i) Make an arrow or a letter "T" shape: only the vertical middle line is a line of symmetry.

(ii) Make a 2 × 8 rectangle: vertical and horizontal middle lines are lines of symmetry.

c **Use these tiles in making creative symmetric designs.**

💡 SAMPLE ANSWER

Make any design and check it by folding or with a mirror; e.g. a square pattern with the same colours in all four corners (4 lines of symmetry).

⭐ Game

? On a 6 × 6 grid, two players take turns covering two adjacent squares with a line. The player who cannot place a line loses. With what strategy can one play to win this game?

✔ ANSWER

Let the other player go first.

After each move, play the **mirror move**: the same line turned by a half turn (180°) about the centre of the grid.

Because the grid is symmetric, this copy place is always free, so you always have a move.

The other player runs out of moves first, so **you win**.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**